

Power BI Lab: Analyzing the Netflix Movies Dataset

Objective:

The goal of this lab is to learn how to import, clean, analyze, and visualize the Netflix Movies dataset using Power BI.

Prerequisites:

- Power BI Desktop installed.
- Netflix Movies dataset (CSV or Excel file).
- Basic knowledge of Power BI functionalities.

Dataset Overview:

The Netflix Movies dataset contains various attributes related to movies available on Netflix, such as:

- **Title:** Movie/TV show title.
 - **Release Year:** Year the movie or TV show was released.
 - **Genre:** Genre(s) of the movie or TV show.
 - **Rating:** Rating of the movie or TV show.
 - **Duration:** Duration of the movie in minutes.
 - **Language:** Language of the movie.
 - **Country:** Country where the movie was produced.
 - **Description:** Short description of the movie or TV show.
-

Lab Steps:

Step 1: Importing the Data

1. **Open Power BI Desktop.**
2. **Click on "Get Data"** from the Home tab.
3. Choose the **"CSV"** option (or **Excel** if your file is in Excel format).
4. Browse for the **Netflix Movies Dataset file** and click **Load**.

5. Power BI will automatically detect the data and load it into the data model.

Step 2: Data Cleaning

1. **Navigate to the "Data" view** by selecting the Data icon in the left pane.
2. Review the dataset and check for missing values, duplicates, or unnecessary columns.
 - To remove duplicates: Go to the "Transform Data" tab, select "Remove duplicates."
 - To handle missing data: Right-click the column, and choose "Replace values" to replace nulls with a placeholder like "Unknown" or "0."
3. **Change data types** if necessary (e.g., ensure "Release Year" is a numeric data type, "Duration" is a whole number, etc.).
4. **Rename columns** for clarity if necessary (e.g., rename "Title" to "Movie Title").

Step 3: Transforming Data (Optional)

1. **Create a calculated column for Movie Length in Hours:**
 - Go to the **"Modeling"** tab.
 - Click **"New Column"** and enter the following formula to convert movie duration from minutes to hours:

Duration Hours = DIVIDE([Duration], 60)

2. **Create a new column for the movie release decade:**
 - In the **Modeling tab**, create a new column with the formula:

Release Decade = FLOOR([Release Year], 10)

- This will group the movies by decade (e.g., 2010s, 2020s).

Step 4: Building Visualizations

1. **Visual 1: Total Movies by Genre:**
 - From the **Visualizations pane**, drag a **Bar Chart** onto the report canvas.
 - Drag the **Genre** field to the **Axis** and the **Movie Title** field to the **Values** (count of movie titles).
 - This chart will show the number of movies available per genre.
2. **Visual 2: Movies Released by Year:**
 - Create a **Line Chart**.
 - Drag **Release Year** to the **Axis** and **Movie Title** (count) to **Values**.
 - This chart will show the trend of movie releases over the years.
3. **Visual 3: Top 10 Movies by Rating:**
 - Create a **Table**.
 - Drag **Movie Title**, **Rating**, and **Duration** onto the values field.

- Apply a **Top N filter** on the **Rating** field to show the top 10 highest-rated movies.
- 4. **Visual 4: Distribution of Movie Duration:**
 - Create a **Histogram** to visualize the distribution of movie durations.
 - Drag **Duration** to the **Values** and choose a suitable bin size (e.g., 10 or 20 minutes).
- 5. **Visual 5: Country-wise Movie Count:**
 - Create a **Map** visualization.
 - Drag **Country** to the **Location** field and **Movie Title** to the **Size** (count of movies).
 - This will show the number of movies produced by country.

Step 5: Creating Insights

1. **Create Insights for Analysis:**
 - Use the “**Card**” visualization to display key metrics such as:
 - **Total number of movies.**
 - **Average movie duration.**
 - **Most common genre.**
 2. **Filter and Slicers:**
 - Add slicers for **Genre**, **Rating**, and **Release Year** to enable users to filter the data interactively.
 - Use the slicers in combination with the visualizations to explore the dataset from different perspectives.
-

Step 6: Formatting & Design

1. **Enhance your visuals** by:
 - Applying appropriate colors.
 - Customizing fonts for titles and labels.
 - Formatting the axes, gridlines, and labels for clarity.
 2. **Use Titles for Context:** Make sure each visualization has a clear and descriptive title to explain what the data represents.
 3. **Align and Resize:** Arrange the visuals on the canvas to avoid clutter and ensure they are easy to interpret.
-

Step 7: Publishing and Sharing

1. **Save your work** and give it a meaningful file name (e.g., "Netflix_Movies_Analysis.pbix").

2. **Publish** the report to the Power BI Service:
 - Click **Publish** from the Home tab.
 - Sign in to your Power BI account and select the workspace.
 3. Once published, you can share the report with others or embed it into a web page.
-

Conclusion:

In this lab, you learned how to:

- Import and clean the Netflix Movies dataset in Power BI.
 - Create meaningful visualizations to explore the data.
 - Use filters, slicers, and measures to generate insights.
 - Format and design your report for professional presentation.
-

Additional Notes:

- For more advanced analyses, consider exploring **DAX (Data Analysis Expressions)** for deeper calculations and **Power Query** transformations for more complex data manipulation.
- Power BI also supports integrating data from other sources, which could be helpful if you're looking to expand the dataset in the future.